

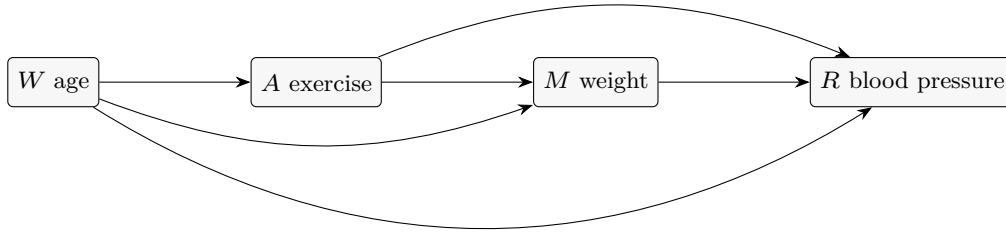
The CDP project explained (v2.2, 4 September 2026) — for the RAI interns

Reading handout for the project. Assumed background: DAGs, the do-operator, back-door adjustment, potential outcomes, regression, a little machine learning. Each component is tagged [NEW] (this project) or [REUSE: source] (existing work the papers apply).

0. What we are trying to draw, and why it is hard

You have a fitted prediction model f (any learner). You want a picture of how its output depends on one input feature A when the features are causally related.

Running example. Features: age W , weekly exercise A , body weight M , blood pressure R ; f predicts 10-year cardiovascular risk. The analyst’s causal graph over the *features* (the “explanatory causal model”, ECM):



The partial dependence plot (PDP). [REUSE: Friedman 2001] For each a , set everyone’s exercise to a , keep their other features as recorded, average the predictions. It silently holds weight and blood pressure fixed, which is not what would happen if someone actually exercised more.

Causal dependence plots (CDPs). [REUSE: Loftus, Bynum & Hansen 2024] Use the ECM to say what “changing A ” means. Four named plots:

plot	question	in the example
TDP (total)	set $A = a$; let everything downstream respond	weight and BP move
NDDP (natural direct)	set $A = a$; hold downstream features at their natural values	this <i>is</i> the PDP (their Theorem 2.10)
NIDP (natural indirect)	keep your own A ; move downstream features to where they’d be under $A = a$	your exercise, but the weight and BP you’d have at exercise a
PCDP (partially controlled)	set $A = a$ and fix another feature	exercise = a and weight = m

The source paper computes all of these by fitting every equation of the ECM and simulating counterfactuals person by person. It draws no sampling-based confidence bands (it can show how the curve varies across candidate graphs, which is a different thing).

The project’s starting fact. [REUSE for the theorem: Pearl 2009, Thm 3.2.2; Zhao & Hastie 2021. NEW: what it implies for CDPs.] Call the model’s output a *constructed outcome*, $Y^* = f(X)$. Then the average TDP is a plain back-door adjustment:

$$\theta(a) = \mathbb{E}[f(X) \mid do(A = a)] = \mathbb{E}_W \left[\mathbb{E}[Y^* \mid A = a, W] \right], \quad W = \text{parents of } A.$$

In the example: regress the model’s prediction on (exercise, age) and average over the age distribution. The equations for weight and blood pressure are never needed, because $do(A = a)$ leaves the parents’ distribution alone and leaves every downstream conditional distribution given (a, w) alone.

A picture that organizes the whole family. [REUSE: Shpitser & Tchetgen Tchetgen 2016] Add the model’s output $\hat{Y}$ to the graph as a child of every feature. Then:

- the TDP and the PCDP are ordinary interventions on *nodes* ($do(A = a)$; $do(A = a, C = c)$);
- the NDDP and the NIDP are interventions on *edges*: the NDDP sets a on the arrow $A \rightarrow \hat{Y}$ and leaves the arrows from A to its child features at their natural value (everything further downstream then follows); the NIDP does the opposite.

Edge interventions in which one node “acts two ways at once” normally need cross-world assumptions that observational data cannot check. Here the node acting two ways is A , but the only conflict that matters for the output is between the arrow into $\hat{Y}$ and the arrows into the features, and $\hat{Y} = f(X)$ is a *known function* — so nothing cross-world is actually used. That is why the NDDP is just an average of a known function (the PDP) and why the NIDP turns out to be an ordinary single-world quantity (next section).

1. Paper A — Reduced-form identification and inference for aggregate CDPs

This is the intern-term paper. It has a sensitivity section (§1.6) that would only become a separate Paper B if a hard piece of theory is finished.

1.1 Every named plot except the PCDP needs only the parents plus one “reduced form” [NEW]

Let R be the features other than A and its parents W , and let $Q(dr \mid a, w)$ be the conditional law of R given (A, W) — a *reduced form*, learnable from data without any equations. Then

plot	population target	what it consumes	statistical character
TDP	$\int P(dw) \int Q(dr \mid a, w) f(a, w, r)$	parents + reduced form	AIPW; $\sqrt{n}$ at discrete levels
NDDP = PDP	$\mathbb{E}[f(a, X_{-A})]$	no causal nuisance at all (only f and the reference distribution of X_{-A})	sample mean of a known function; $\sqrt{n}$ at each a
NIDP	$\int P(da', dw) \int Q(dr \mid a, w) f(a', w, r)$	parents + reduced form + the law of A given W	its own influence function; $\sqrt{n}$ at discrete levels
PCDP	needs the graph <i>inside</i> R	more than the reduced form (two graphs with the same data give different answers)	g-formula

The NIDP line is reuse, not a discovery. [REUSE: Fulcher, Shpitser, Marealle & Tchetgen Tchetgen 2020, *JRSSB*] The quantity “your own A , but the downstream values you would have had under $A = a$ ” is the *bridge mean* $\Psi(a) = \mathbb{E}[Y(A, Z(a))]$ of their population-intervention indirect effect (the effect itself is the contrast $\mathbb{E}(Y) - \Psi(a)$), with

Z = the downstream features and $Y = f$. Their Lemma 1 gives the formula above, their Theorem 1 gives its influence function (one term drops out because Y^* is deterministic), and their doubly robust estimator applies: it is consistent if *either* the law of A given W *or* the reduced form is estimated correctly. What is new here is recognizing that the published NIDP *is* this object, that the covariate set can be reduced to the parents of A , and the simplifications that follow from f being known. In this term’s paper the NIDP is handled for discrete A only; its continuous-feature version needs its own theory.

Direct + indirect $\neq$ total. [NEW, modest] With B the average prediction,

$$\theta(a) - B = [\text{NDDP}(a) - B] + [\text{NIDP}(a) - B] + J(a),$$

where J is an *interaction remainder* (it vanishes when f is additive in a and the downstream features). A tiny example: $A \sim \text{Bern}(p)$, downstream $R(a) = a$, $f = ar$; the total contrast is 1, the direct and indirect contrasts are each p . So the two named plots do not add up to the total in general; J can be plotted alongside them.

1.2 Estimating one level when A is discrete [REUSE]

Three classical estimators [REUSE: Robins, Rotnitzky & Zhao 1994; Hahn 1998]: regress Y^* on (A, Z) and average; weight units with $A_i = a$ by $1/\hat{\pi}(a | Z_i)$; or **AIPW**, which combines them and is *doubly robust* (right if either piece is right):

$$\hat{\theta}(a) = \frac{1}{n} \sum_i \left[\frac{\mathbf{1}(A_i = a)}{\hat{\pi}(a | Z_i)} (Y_i^* - \hat{\mu}(a, Z_i)) + \hat{\mu}(a, Z_i) \right].$$

With **cross-fitting** [REUSE: Chernozhukov et al. 2018] — fit $\hat{\mu}, \hat{\pi}$ on one half, evaluate on the other, swap — you may use flexible ML for the nuisances and still get a valid interval $\hat{\theta}(a) \pm 1.96 \hat{\text{se}}$, provided the two nuisance errors shrink fast enough *jointly* (their product must be $o_p(n^{-1/2})$). Cross-fitting is not magic: it removes a technical condition, not the accuracy requirement.

1.3 Which covariates Z to adjust for [REUSE: Rotnitzky & Smucler 2020; NEW application]

The parents W are *valid* but need not give the smallest variance. Adding non-descendants of A that help predict Y^* lowers the variance without changing the answer — the same reason good covariates help in a randomized trial. When f uses all features, the set of **all non-descendants of A** is never worse than any other valid set in large samples, among the adjustment estimators we use (equality is possible: an added variable need not carry new information). In finite samples the bigger regression can behave worse because it is harder to fit — one reason the comparison study exists. Two facts the interns should know: (a) the *population* positivity condition does not get harder, because $P(a | \text{non-descendants}) = P(a | W)$ by the graph; what gets harder is fitting a bigger regression — so fit the propensity on W and the outcome regression on the large set; (b) this uses the graph’s arrows, never its equations. (Verified exhaustively on small graphs in the review.)

1.4 Continuous A : bins first, curves second [REUSE throughout; D4]

- **Bins (primary).** At each bin B_k report the fixed-weight average $\psi_k = \int_{B_k} \theta(a)g_k(a) da$ with a weight chosen *in advance* (uniform, say — weighting by the observed distribution of A inside the bin is possible but makes the target depend on the data law and adds a term to the standard error). It is a regular, $\sqrt{n}$ target estimated by the same AIPW with weight $g_k(A)/\hat{\pi}(A | Z)$, and it does not depend on which valid Z you use. (The alternative “natural” within-bin policy is also legitimate but its definition includes the conditioning set; appendix material. The two bin targets are different numbers in general.)
- **Smoothed curve (extension, supervisor-owned).** $\theta_h = K_h * \theta$ at a declared bandwidth h , estimated by a kernel-weighted AIPW, bands by multiplier bootstrap over a grid. Like the bins, it works for *any* f — important, because a tree that splits on A may make θ itself jumpy in a , and the smooth-curve theory below then does not apply.
- **Smooth curve (secondary).** If θ is smooth, one published engine, cited as the engine: Takatsu & Westling’s debiased local-linear estimator with uniform bands, or Semenova & Chernozhukov’s cross-fitted series estimator (whose variance has an extra term that must not be dropped). The pseudo-outcome must be Kennedy’s [REUSE: Kennedy et al. 2017]: the naive “ $\hat{\mu} + (Y^* - \hat{\mu})/\hat{\pi}$ regressed on A ” estimates the wrong thing (checked: off by 0.9 in a toy example).
- **Bias policy.** [NEW correction] Any uncorrected smoother at its best bandwidth has bias as large as its standard error, so naive bands under-cover; the project’s earlier prototype had exactly this problem (0.89 coverage). Use debiasing, deliberate undersmoothing, or report bands for θ_h .

1.5 Adjustment versus propagation [NEW study]

Four arms: parents only; the efficient set; an estimator that exploits the DAG’s factorization; and full parametric propagation (fit every equation, simulate). What the study should show, precisely:

- A confounder that lives entirely downstream (say between weight and blood pressure, independent of exercise) does not invalidate the parent-adjustment formula for the *total* curve: the formula is right whether or not that confounder is drawn in the graph. (It does change the data, and so the curve; the point is that adjustment still finds the right curve.) Adjustment inherits this automatically. Propagation inherits it too *if* its fitted conditionals match the observational ones — exactly so for linear OLS (an algebraic identity: fitted path coefficients always compose to the reduced-form slope) — but a downstream confounder changes the *shape* of those conditionals, so a Gaussian propagation model can have the right mean and variance and still get Ef wrong for a nonlinear f (checked: error 0.047 with mean and variance exact).
- When the ECM’s parametric equations are correct, propagation can be more precise and can extrapolate outside the region where data overlap — as model-based extrapolation, declared as such. Overlap is diagnosed with the non-overlap ratio [REUSE: Bao & Schomaker 2026].
- Whether exploiting the factorization can beat adjustment at all can be read off the

graph [REUSE: Rotnitzky & Smucler 2020, Thm 19].

1.6 Sensitivity section [mostly REUSE; NEW framing]

If an unmeasured U affects A and a downstream feature that matters for f , it can bias the adjusted curve (in special cases the bias cancels or U is irrelevant to f). Partial- R^2 sensitivity [REUSE: Cinelli & Hazlett 2020; nonparametric bounds: Chernozhukov, Cinelli, Newey, Sharma & Syrgkanis] asks how strong U would have to be to flip the *slope* (never the level — that depends on an arbitrary zero), flatten the curve, or cross a threshold. Project-specific content: confounding parameterized per ECM edge, with an exact map to the standard coordinates (the closed-form “tipping strength” is the Cinelli–Hazlett robustness-value formula in edge coordinates); bounds at each bin; benchmarking against observed predictors; and the invariance contrast of §1.5. Existing work on sensitivity for dose-response curves [REUSE: Bonvini et al. 2022; Marmarelis et al. 2023; Jesson et al. 2022; Baitairian et al. 2025 (sharp bounds, arXiv 2411.02231; and their calibration review, arXiv 2510.16560)] can be instantiated with $Y^* = f(X)$ whenever its support, moment, regularity, smoothness, and sampling assumptions hold, and is cited as such.

1.7 Two practical rules [NEW, cheap, important]

- **Explanatory data must be held out.** $Y_i^* = f(X_i)$ on rows used to *train* f is not a fair sample of what f does (a forest nearly reproduces its training outcomes). Draw every curve on rows the model never saw. Out-of-bag or cross-fitted predictions are predictions of *different* models, not of f .
- **Name the reference population.** The outer average is over some distribution of W ; training cohort, held-out sample, and deployment population give different curves whenever the effect varies with w . All the inference in this term’s paper is for the held-out sample’s population; carrying a curve to a different population (“transport”) needs extra data and different formulas and is not in scope.

1.8 What Paper A does not claim

That parent adjustment, AIPW, cross-fitting, or dose-response bands are new; that the method works “without a causal model”; that it is the first semiparametric inference for causal explainability [REUSE: Zhang & Gao 2026; Gao & Zhao 2024, different quantities]; bands for the individual per-person curves (out of scope).

1.9 Feasibility and supervision (matches PLAN_v3.md §10)

task	intern-feasible?	supervision
Verify §1.1 by exact enumeration in toy SCMs (review scripts are templates)	yes	light
Cross-fitted AIPW at discrete levels; coverage on the oracle testbed with Monte-Carlo tolerances	yes	moderate
Parents vs efficient set (variance drop; propensity on parents)	yes	moderate

task	intern-feasible?	supervision
Held-out explanatory sample; reference population; overlap diagnostic; interaction plot	yes	light
Fixed-bin targets with multiplier-bootstrap bands	yes	moderate (representer, bootstrap over a finite vector)
Smoothed curve θ_h	no	supervisor-owned extension
Propagation arm via DoWhy; invariance demonstrations	yes	moderate
Robustness section: linear closed forms, planted confounders, tipping, benchmarking	yes	moderate; binned Riesz bounds only if time allows
Smooth-curve secondary results; efficiency theory; writing the “what is new” framing	no / with help	supervisor

2. Paper C — exact semantics and graph uncertainty (later project, not for this term)

A theory paper that makes the picture of §0 rigorous: a lemma saying why a deterministic output node removes the cross-world content of edge interventions; influence functions for all four plots (the NIDP’s is Fulcher’s; the PCDP’s is a g-formula one); efficiency when the whole DAG is used [REUSE: Rotnitzky & Smucler 2020 §6; Guo, Perković & Rotnitzky 2023]; and a section on **graph uncertainty**: from observational data you often cannot orient every edge, so there are several candidate parent sets and several candidate curves [REUSE: Maathuis, Kalisch & Bühlmann 2009; Perković 2020; Guo & Perković 2021; Chang, Guo & Malinsky 2025; Edgar & Battey 2026]. Possibly new there: curve-valued union bands, and merging candidates that give the same curve *for this f* . Caution: for linear-Gaussian models all graphs in an equivalence class have identical likelihood, so a BIC “neighbourhood” of orientations is empty there; for nonlinear models orientations are scored. Intern-sized pieces exist (simulating candidate curves with IDA software; the merging algorithm) but the theory is research-level.
